UNIVERSITY OF LONDON

BSc EXAMINATION 2019

For Internal Students of
Royal Holloway

DO NOT TURN OVER UNTIL TOLD TO BEGIN

CS3003: IT Project Management

Marking Scheme

Time Allowed: **TWO** hours

Answer THREE questions
No credit will be given for attempting any further questions

Calculators are NOT permitted

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General Note

The exam questions are structured in such a way that approximately 50% of the marks come through bookwork, while for the remainder students need to apply the knowledge, analytical and judgment skills gained in the course to make decisions or solve new problems.

Summary of modules' learning outcomes

LO1 - Understand the fundamental concepts, terminology and cost elements of IT Projects.

LO2 - Apply basic Project Planning techniques.

LO3 - Explain the procedures needed to monitor, control and report upon an IT Project.

LO4 - Demonstrate an understanding of the steps needed to build and maintain effective development teams.

LO5 - Discuss and where appropriate apply the principles of Project Risk Management.

LO6 - Explain the ways in which appropriate quality attributes of the products of an IT Project can be assessed and assured.

Mapping of Exam Questions to Learning Outcomes

Question Number	Learning Outcome
1a	*LO1*
1b	*LO6*
1c	*LO3, LO5*
2a	*LO1, LO2*
2b	*LO2*
2c	*LO1*
3a	*LO1*
3b	*LO3*
3c	*LO5*
3d	*LO1, LO2*
4a	*LO4*
4b	*LO1*
4c	*LO1, LO3*
4d	*LO3, LO6*

1.

(a) A company is about to buy a software package, Package A, that consists of three components, namely A1, A2 and A3. These components are licensed independently according to the following scheme:

- A1 is licensed on a per user basis, costing £100 per user that will have access to the package.
- A2 is licensed on a per user basis, costing £300 per user that will have access to the package.
- A3 is licensed based on the number of CPU's (processors) of the server that Package A will be installed. The license cost of A3 is £20,000 per installed CPU.

  It is estimated that in order to adequately perform in the production environment, Package A requires 4 CPUs for up to 250 users, 8 CPUs for 251 to 500 users and 16 CPUs for 501 to 1000 users.

Moreover, starting from the second year, the company will have pay the vendor of Package A an annual maintenance fee of 20% over the license fee. Finally, each CPU of the production environment is acquired at a cost of £10,000 and has an annual maintenance fee of 10% over the acquisition cost. The CPU maintenance fee also starts from the second year.

If variable N denotes the number of users and variable M the number of CPUs, based on the previous facts, perform the following tasks:

(i) Write down the formulas to calculate the 5-year Total Cost of Ownership (TCO) of the investment that the company has to make for Package A; [10 marks]

(ii) Calculate the 5-year TCO of Product A for 300 users. [6 marks]

Notes: There is no disaster site for Product A. The time-value of money should not be taken into account in the TCO calculation.

Marking Scheme (*non-bookwork*)

4 marks for correct licensing calculation, 4 marks for S/W annual maintenance, 2 marks for H/W acquisition and maintenance, 6 marks for TCO calculation for 300 users.

(b) In the context of project quality, explain the main differences between quality assurance and project assurance. [6 marks]

Marking Scheme (*bookwork*)

4 marks for adequate description of the difference in scope. 2 marks for the fact that quality assurance is independent of the project and that project assurance is independent of the project manager.

(c) Briefly explain the two main strategies that may be applied to rollout an IT system. Discuss two risks of each strategy.

[12 marks]

Marking Scheme (*partially bookwork*)

2 marks for each rollout strategy correctly identified, 2 marks per risk for each rollout strategy.

2.

(a) Indicate whether each of the following statements is **true** or **false**. A justification is **not** required.

(i) Work Breakdown Structures (WBS) are beneficial because they assist in managing the timing aspects of a project;

(ii) Work Breakdown Structures (WBS) can be used as input in determining the cost and budget of a project;

(iii) A WBS dictionary is a document that describes technical terms used for scope management;

(iv) Change requests are a sign of poor planning and should always be avoided;

(v) Parametric estimation methods depend on the existence of a historic model.

[10 marks]

Marking Scheme *(partially bookwork, students should demonstrate appropriate understanding of bookwork).* **2 marks per correct answer**

(b) The activities of a project and respective durations and dependencies are shown in the following table.

Activity	Duration (weeks)	Predecessors
A	4	-
B	3	A
C	2	B
D	1	C
E	2	B
F	8	E
G	10	D
H	3	B
I	3	B
J	4	F,G,I
K	3	H
L	3	J,K
M	1	L

(i) Draw a network diagram for this project including all floats; [11 marks]

(ii) Mark the project's critical path on the diagram. [4 marks]

Marking Scheme (*non-bookwork*)

i. **1 mark for each correct node/sequence/float in the diagram, excluding nodes A and M. Deduction of 2 marks for each wrong node/float.**

ii. **Full marks for correct critical path**

(c) In the context of cost estimation, provide three potential issues or disadvantages of applying the Delphi method. [9 marks]

Marking Scheme (*non-bookwork*)

3 marks for each issue or disadvantage that makes sense

3.

(a) Consider the following stages of an IT Project: Business Case, Requirements Analysis, System Design, Testing, post-rollout Operation. Complete the table below, indicating, on each of these stages, whether the tasks of an off-the-shelf package-based implementation would be *similar or different* compared to a custom-built software development approach. Justify your answer by referring to elements of convergence or divergence between the two approaches. One table row has been completed for you as a sample answer.

Project Stage	Package-based Vs Custom-built	Justification
Business Case	*Similar*	*In both approaches a business case should be developed, including elements such as strategic drivers, objectives, costs, benefits and risk of the project.*
Requirements Analysis		
System Design		
Testing		
post-rollout Operation		

[12 marks]

Marking Scheme (*non-bookwork*)

3 marks for each correctly identified similarity per project stage with appropriate justification. 1 mark for correct similarity without justification. An answer that the similarity is not appropriately identified but a sound justification is provided will be considered as correct.

(b) Indicate four different types of project meetings. For each type, outline its objective, typical agenda and participants. [8 marks]

Marking Scheme (*bookwork*)

2 marks per type of project meeting, including correct objective, agenda and participants. 1 mark for merely an identification of a meeting type.

(c) Explain the notion of a project's Risk Management Plan and indicate four items that such a plan should contain. [6 marks]

Marking Scheme (*bookwork*)

2 marks for correct definition of risk management plan. 1 mark for each correct type of item that such a plan should contain.

(d) In the context of stakeholder analysis, sketch a 2x2 Power/Interest grid. For each of the four categories in the grid, outline a suitable approach that you might take in order to manage stakeholders. [8 marks]

Marking Scheme (*partially bookwork*)

4 marks for correct and complete grid. 1 mark for appropriate approach in each of the four categories. Proportional marks otherwise (e.g. 2 correct categories in grid will provide 2 marks).

4.

(a) Two software developers of your team are involved in a serious conflict. It is apparent that the conflict is based on different interpretations of the project's scope. Which of the following resolution approaches would you apply in this case? Briefly **justify** your answer.

- Compromise
- Accept/Do nothing
- Impose a judgement
- Defuse

[8 marks]

Marking Scheme (*non-bookwork*)

8 marks for correct answer appropriately justified. 4 marks for incorrect answer with a good justification.

(b) Your organization has approved a project for a new Enterprise Resource Planning (ERP) system. There exist two implementation options: buying a packaged solution or developing the new system in-house. Provide five factors you would consider in deciding which option to take. [10 marks]

Marking Scheme (*non-bookwork*)

2 marks for each factor that makes sense.

(c) A project has a Cost Performance Index (CPI) of 0.90 and a Schedule Performance Index (SPI) of 0.95. Which of the following statements is true? Briefly **justify** your answer.

- 95% of the whole project has been completed and 90% of the project's overall budget has been spent;
- 95% of the whole project has been completed and 90% of the project's budget planned to date has been spent;
- 95% of the work planned to date has been completed and 90% of the project's budget planned to date has been spent;
- 95% of the work planned to date has been completed and 90% of the project's overall budget has been spent.

[7 marks]

Marking Scheme (*non-bookwork*)

7 marks for correct answer. No marks otherwise (the question tests whether students have correctly understood the concepts of a project's CPI and SPI)

(d) In the context of Configuration Management, explain the purpose of each of the following activities:

(i) Change Management;

(ii) Version Management;

(iii) Release Management.

[9 marks]

Marking Scheme (*bookwork*)

3 marks for each correct answer in each of the activities above.